C-EX: evidence for a weak cadmium-associated reporter response

Which explanations are assessed by the supplied data, and which need a different measurement?

Materials and methods

F01_expanded_summary.json: frozen task summary or dated source-context review.

CEX_C06_summary.json: frozen task summary or dated source-context review.

F01_context_review.json: frozen task summary or dated source-context review.

Preserve source result sentences verbatim; classify evidence type without scores or statistical tests.

Compare the measured scope with the measurement each causal explanation requires.

Scope and working assumptions

F-01 acquisition and nominal-condition assumptions are inherited; C-06 is reference-only. This is a completed evidence review, not a causal expression, toxicity or direct-binding diagnosis.

Results

The review retains 7 F-01 findings and 4 C-06 findings verbatim in source_findings.csv.

Recorded imaging findings from F-01: 13 October, nominal 10 mM Ca / 5 mM Cd: C8: +2.66 ADU over 298.2s; Empty: +0.12 ADU over 299.5s; C6: +15.01 ADU over 298.4s; C1: +0.57 ADU over 299.6s; C2: +0.77 ADU over 299.6s. The C6 recording has the largest within-recording increase in this group. Candidate field-matched pre/post contrasts are C1 -10.22 ADU; C2 +1.92 ADU; Empty +0.68 ADU. Within-recording change and the field-matched treatment contrast answer different questions: the C1 field decreases across its paired acquisitions despite rising slightly within the challenged recording.

Selected reference-only finding from C-06:

The labelled CDS spans reference bases 748-2097: 1350 nt and 450 aa; its translation matches its annotation.

Seven questions are classified by evidence type; direct expression, biological power and direct binding remain unresolved.

Question	Evidence type	Input data	Finished without more wet-lab data?
Recorded reporter behavior	Measured imaging	F01_expanded_summary.json	Yes: evidence review; no: causal diagnosis
Optical artifacts / background	Measured imaging	F01_expanded_summary.json	Yes: evidence review; no: causal diagnosis
Reference codon composition	Reference sequence	CEX_C06_summary.json	Yes: evidence review; no: causal diagnosis
Low protein expression is the cause	Unresolved	No direct expression data	Yes: evidence review; no: causal diagnosis
Growth / viability / toxicity	Preparation context	F01_context_review.json	Yes: evidence review; no: causal diagnosis
Insufficient biological power	Unresolved	F01_expanded_summary.json	Yes: evidence review; no: causal diagnosis
Direct Cd binding or another site	Unresolved	No binding data	Yes: evidence review; no: causal diagnosis

Interpretation

The available imaging and reference analyses support a structured assessment of reporter behavior, technical conditions and reference codon composition. Low protein expression has not been identified as the cause of a weak response. Matched protein abundance, independent-culture identities and same-sample growth/viability measurements distinguish the remaining explanations.

References and reproducibility

F-01 and C-06 frozen summaries: primary methods and references are preserved in those input JSON files.

13 October notebook context: CTX02-CTX06; original source hashes and locators in F01_context_review.json.

Figure 1

Available evidence for the competing explanations

Question / explanation	Evidence category
Recorded reporter behavior	Measured imaging
Optical artifacts / background	Measured imaging
Reference codon composition	Reference sequence
Low protein expression is the cause	Unresolved
Growth / viability / toxicity	Preparation context
Insufficient biological power	Unresolved
Direct Cd binding or another site	Unresolved

Categories describe evidence type, not probability, effect size or independent biological n.

Categorical evidence coverage for seven questions. Colors encode source type only; no probability or statistical effect is implied.